

State Machine Question (100 marks)

Given the problem description below, specify the *state machine* for the automatic bread maker. **Show the states as well as transitions including events, guards, and actions.** You do not need to specify parameters for events, guards, and actions. If you use private methods for guards or actions in your state machine, use meaningful names for those methods. You do not need to define the method bodies of those methods.

Problem Description: Automatic Bread Maker

The automatic bread maker has six buttons (each of which corresponds to one event): **menu**, **crust**, **plus**, **minus**, **start**, and **stop**. Furthermore, the bread maker can be turned **on** and **off**. The display of the bread maker is out of scope for this state machine.

After turning on the bread maker, a user can select the desired course by iteratively pressing the **menu** button. The first course is selected by default. At the same time, the user can select the desired kind of crust. The default is medium, which can be changed to dark and then light and then back to medium by pressing the **crust** button. The bread maker has a delayed start feature, which can be enabled by setting the timer with the **plus** and **minus** buttons. Initially, the delayed start feature is not enabled. Each time the **plus** button is pressed, the timer is increased by 10 minutes up to a maximum of 18 hours. The **minus** button decreases the timer by 10 minutes.

The **start** button initiates the baking process or the countdown of the timer in case a timer is set for a delayed start. The baking process consists of the kneading, rising, baking, and warming phases. The various courses of the bread maker differ in terms of their kneading, rising, and baking times. For the kneading phase, a motor is started with the internal **startKneading()** command. The motor is stopped with the internal **stopKneading()** command at the end of the kneading phase. During the rising phase, the bread maker does not do anything to the dough. The baking temperature is controlled by the internal **heat()** command during the baking phase. Each course has a 60 minute warming phase that starts after the baking phase and keeps the bread maker at a set temperature with the internal **warm()** command. At the end of the warming phase, the bread maker automatically returns to the course selection stage. At any time during the baking process or during the countdown, the **stop** button also returns the user to the course selection stage.